

Counting (Part II) Homework

① Michelle will be on vacation for one week. There are 10 restaurants in the area that she is interested in, 6 spots to snorkel, and 12 shops that she would like to visit. If she has time for only 7 restaurants, 3 spots to snorkel, and 5 shops to visit, how many possible outcome variations are there for her vacation?

② Bill is moving into his new office. He would like to have pictures of his 4 family members. At home he has 15 pictures of his wife, 8 pictures of his son, 10 pictures of his eldest daughter, and 12 pictures of his youngest daughter. He would like to have 2 pictures of his wife at his office, 3 pictures of his son, 4 pictures of his eldest daughter, and 4 pictures of his youngest daughter. How many outcome variations are there for the pictures that he will have in his office?

③ The United Nations will choose a delegation of people from 5 different countries to discuss current events. The organization is considering 9 people from Germany, 5 people from Portugal, 10 people from Italy, 6 people from Britain, and 7 people from Australia. If only 3 people from Germany will be chosen, 2 people from Portugal, 5 people from Italy, 4 people from Britain, and 2 people from Australia, how many outcome variations are possible for the delegation?

④ Let $A = \{-10, -4, 0, 4, 6\}$ and $B = \{-12, 0, 4, 6\}$.
Find $A \cap B$

⑤ Let $A = \{2, 7, 5, 6\}$ and $B = \{-4, 0, 1, 3, 4\}$.
Find $A \cap B$

⑥ Let $A = \{-2, -1, 2, 5, 7\}$ and $B = \{5, -2, -1, 9, 7, 2\}$.
Find $A \cap B$

⑦ Let $A = \{26, 20, 19, 10, 17\}$ and $B = \{26, 19, 17\}$.
Find $A \cup B$

⑧ Let $A = \{-601, -523, 0, 124\}$ and $B = \{124, 127\}$

Find $A \cup B$

⑨ Let $A = \{1, 6, 7\}$ and $B = \{-16, 6, 1, 3, 7\}$.

Find $A \cup B$

Use Venn diagrams to solve the following problems.

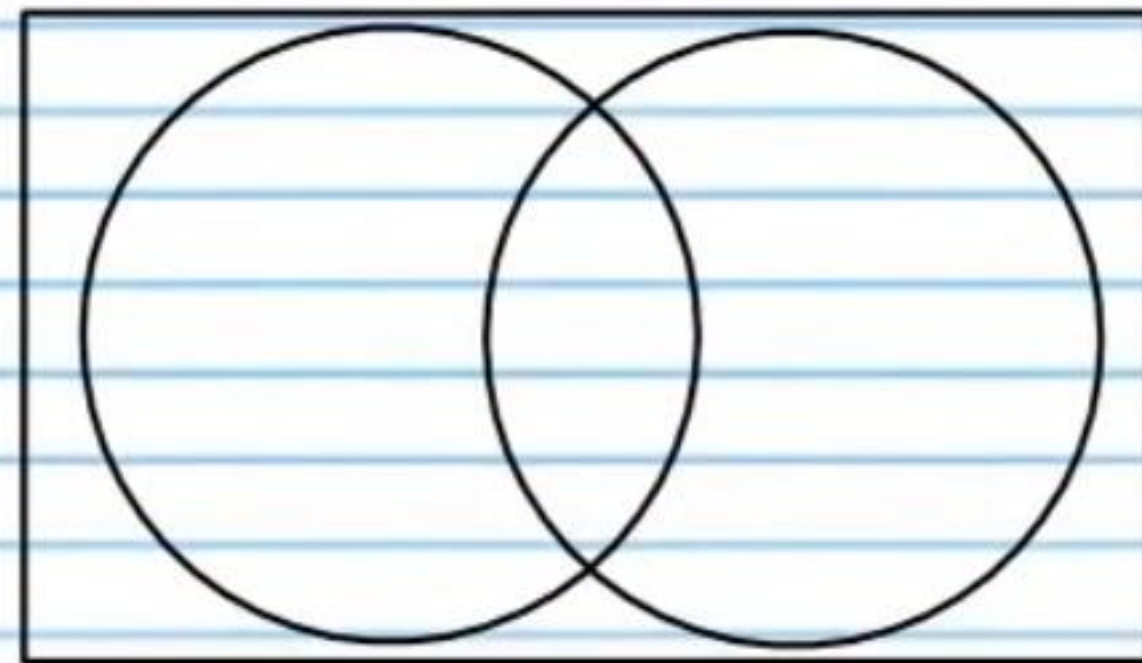
⑩ There is a cluster of 61 trees on a hillside. There are 46 trees with green leaves, 22 trees with yellow leaves, and 14 trees with both green and yellow leaves.

i) How many trees have yellow leaves but no green leaves?

ii) How many trees have green leaves but no yellow leaves?

iii) How many trees have either green or yellow leaves?

iv) How many trees have neither green leaves nor yellow leaves?



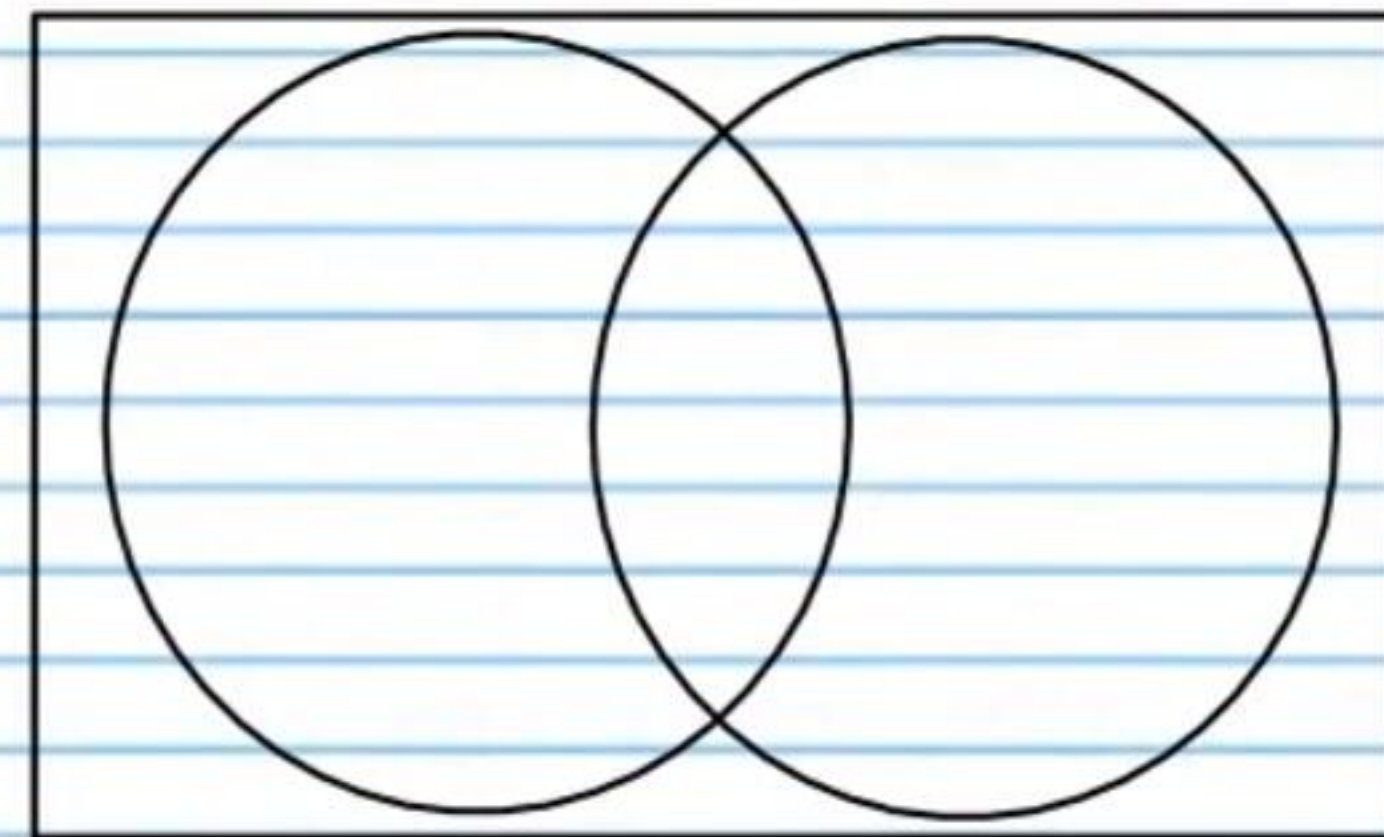
⑪ A bicycle factory made 38 frames in the month of June. 26 of them were made with carbon. 13 of them were made with iron. 7 of them were made with both (a carbon-iron alloy).

i) How many frames had carbon only?

ii) How many frames had iron only?

iii) How many frames had either carbon or iron?

iv) How many frames had neither iron nor carbon?



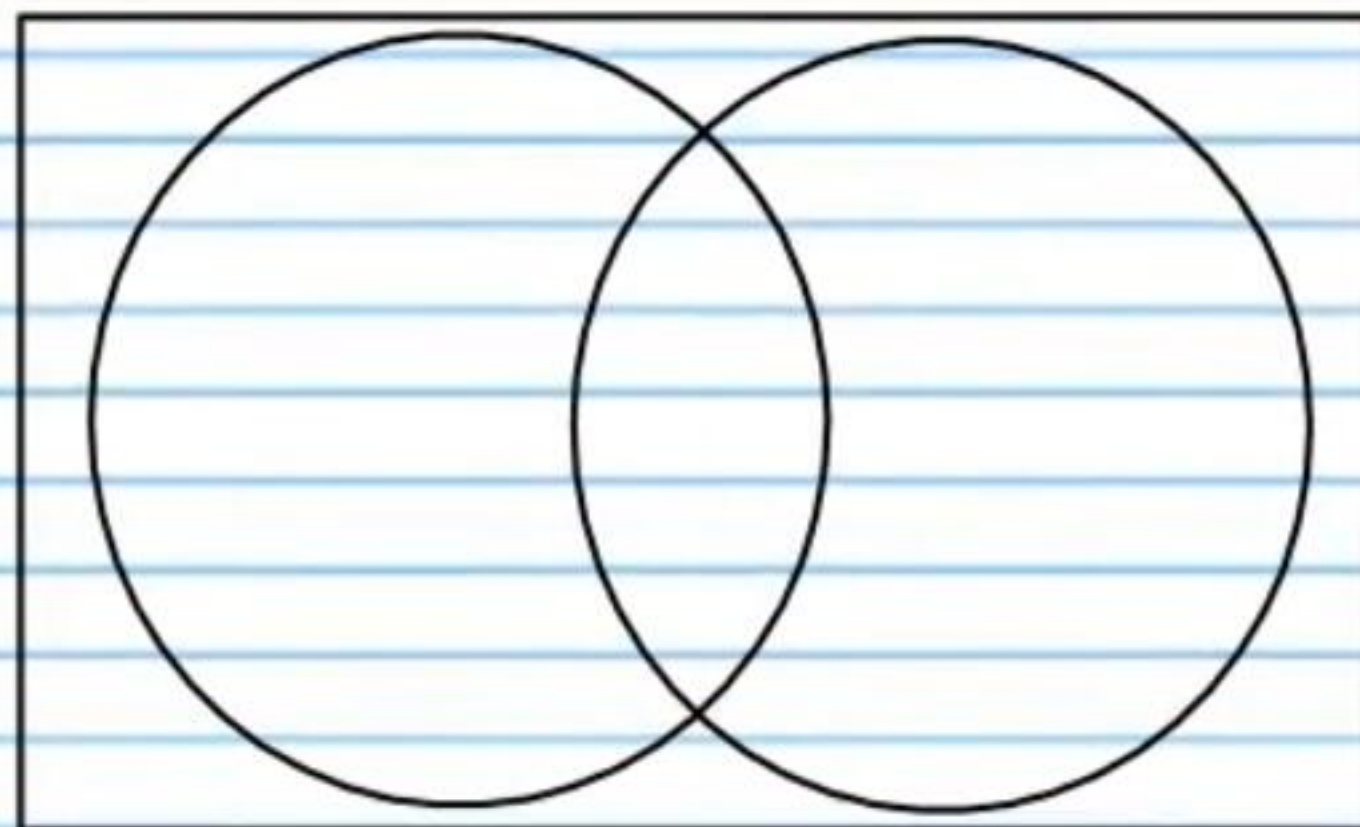
⑫ There are 56 children in a scout troop. 32 of them have rock climbing experience, and 25 of them have knot tying experience. 15 of them have experience with both.

i) How many children have experience with rock climbing but not knot tying?

ii) How many children have experience with knot tying but not rock climbing?

iii) How many children have experience with either knot tying or rock climbing?

iv) How many students have experience with neither rock climbing nor knot tying?



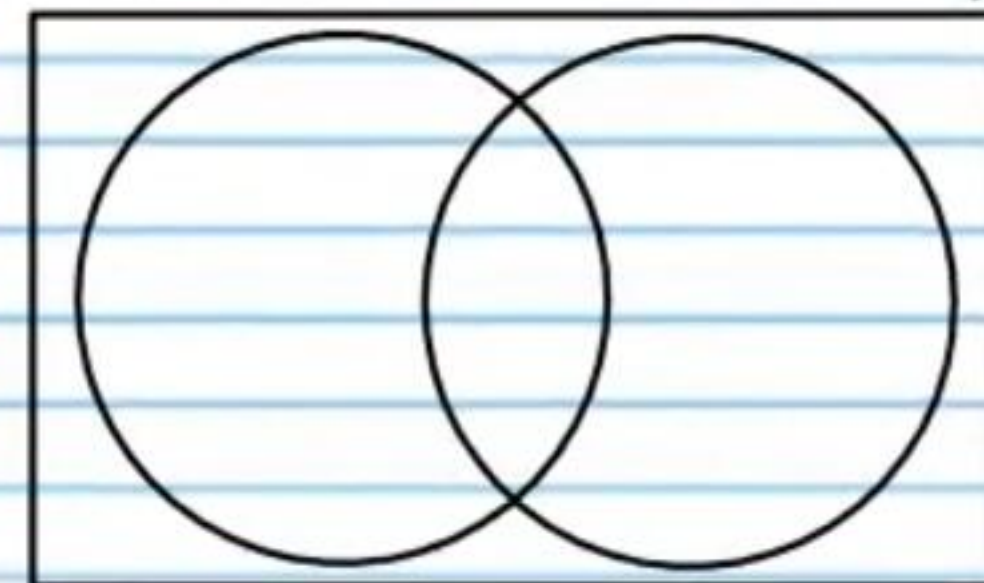
⑬ 22 children collect baseball cards, and 37 children collect football cards. 41 children collect either baseball cards or football cards. There are 94 children in total.

i) How many of these children collect neither baseball nor football cards?

ii) How many of these children collect both baseball and football cards?

iii) How many of these children collect baseball cards only?

iv) How many of these children collect football cards only?



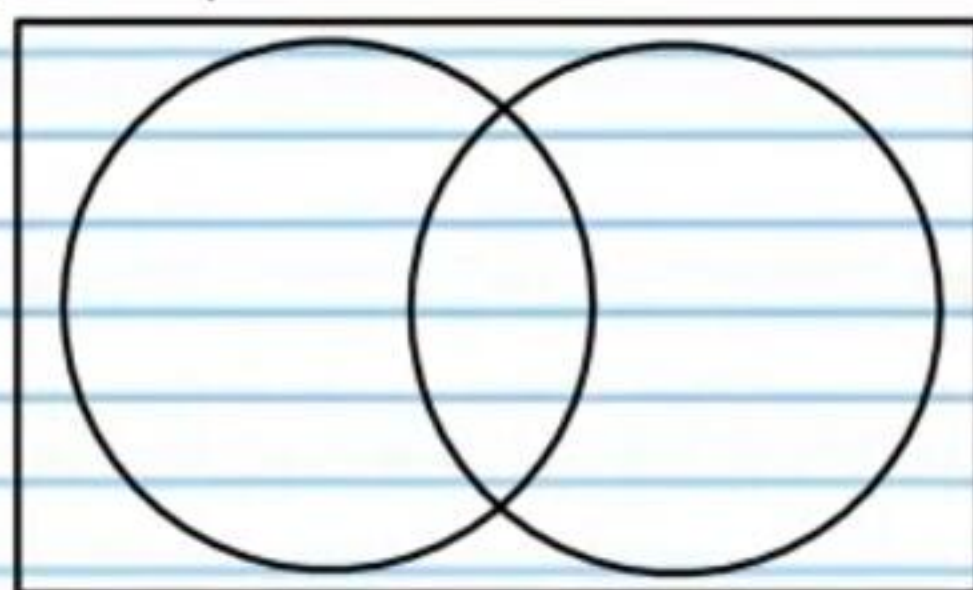
⑭ There are 63 people in an office building. 48 of them speak either Japanese or Mandarin. 30 of them speak Japanese and 25 of them speak Mandarin.

i) How many of the people speak neither Japanese nor Mandarin?

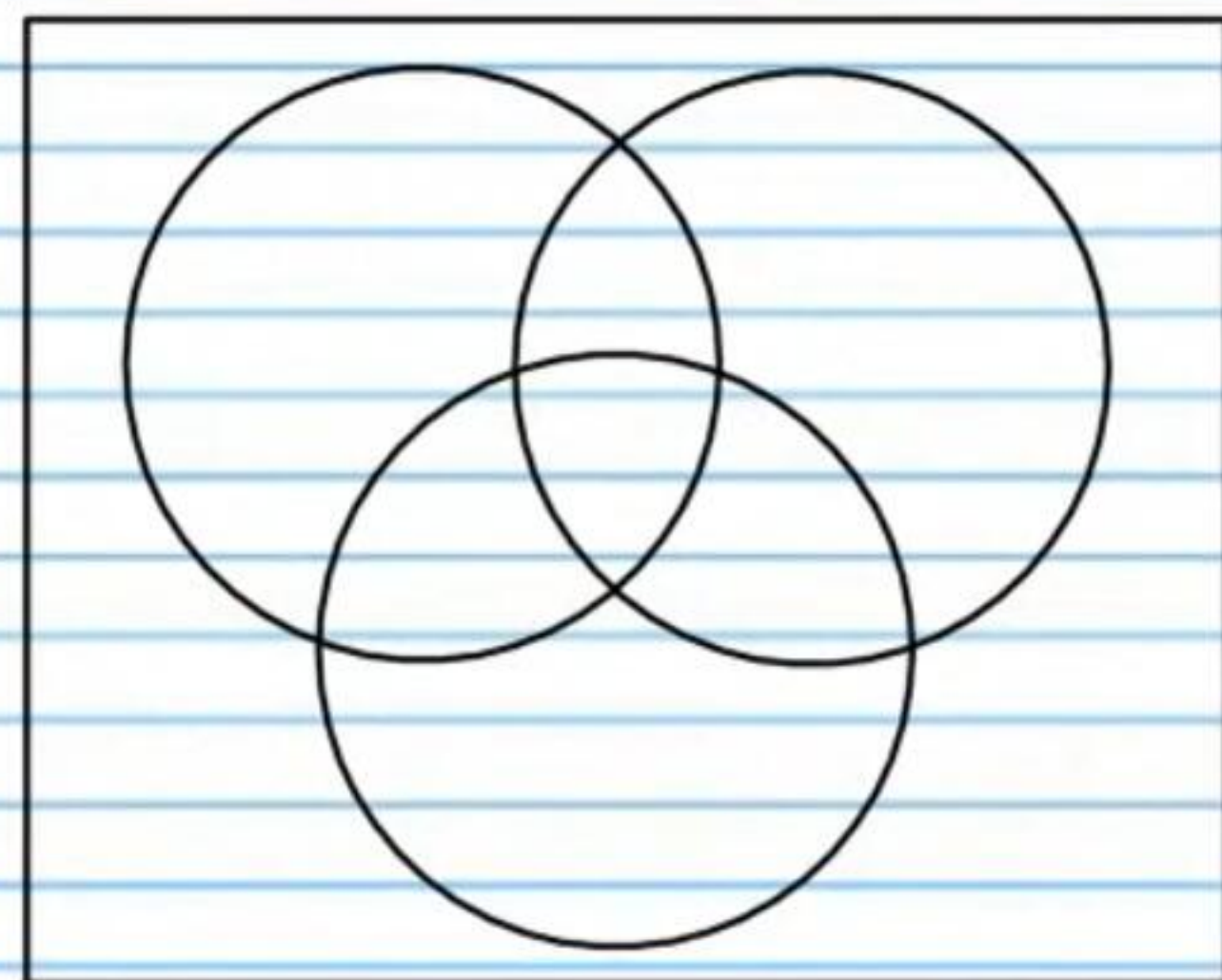
ii) How many of the people speak both Japanese and Mandarin?

iii) How many of these people speak Japanese but not Mandarin?

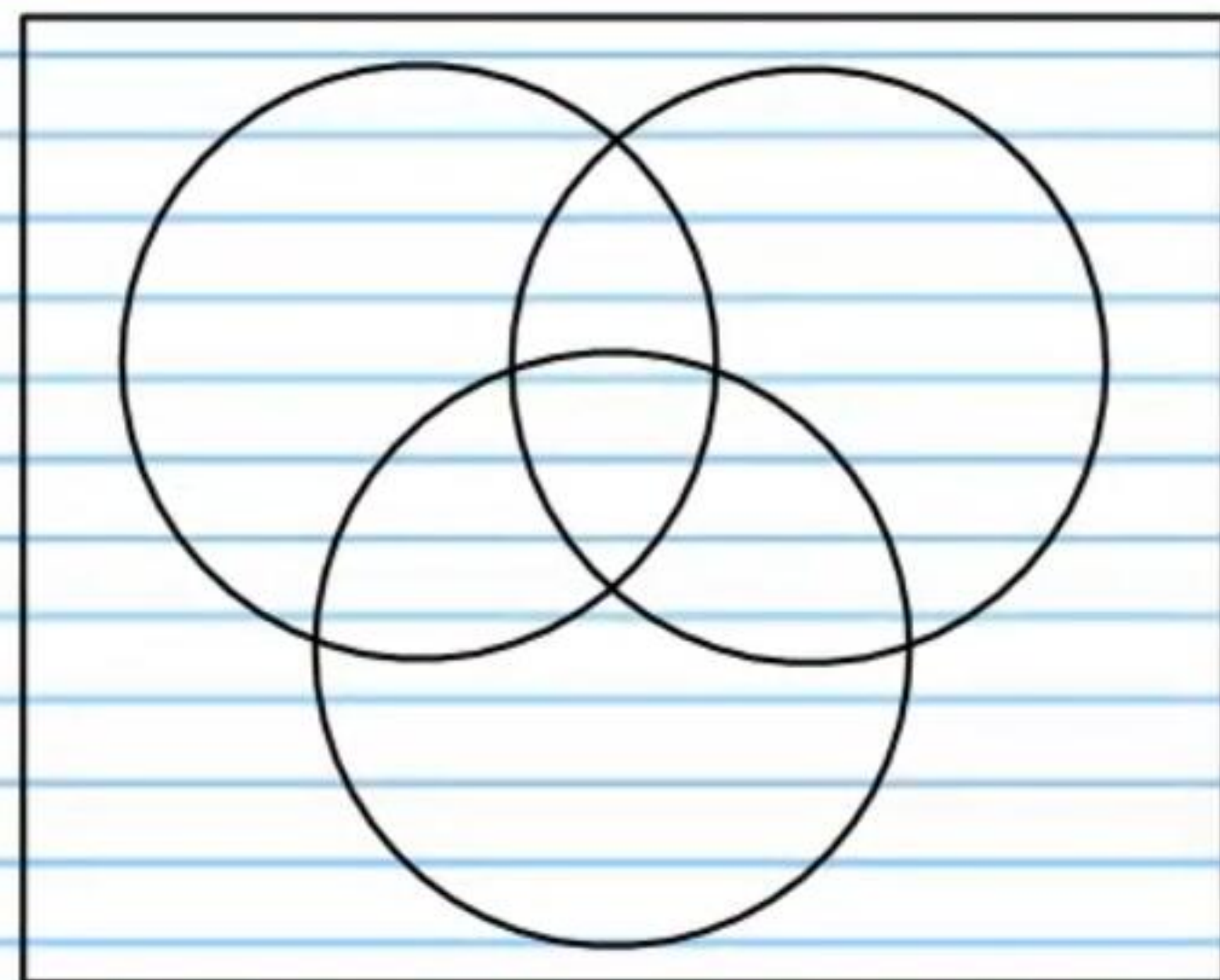
iv) How many of these people speak Mandarin but do not speak Japanese?



⑮ There are 59 restaurants in Paradise City. 28 of them serve baked potatoes, 36 of them serve hamburgers, and 27 of them serve tacos and hamburgers. 21 of them serve tacos and baked potatoes, 15 of them serve baked potatoes and hamburgers, and 10 of them serve all three of these dishes. 6 of them sell none of these dishes. How many restaurants serve tacos, but not baked potatoes or hamburgers?



⑩ There are 37 powersports dealerships in Oregon State. 18 of them sell the Kawasaki brand, 20 sell the Honda brand, and 11 sell the Suzuki and Honda brands. 9 sell the Suzuki and Kawasaki brands, and 7 sell the Kawasaki and Honda brands. 6 sell all three of these brands and 3 of them sell none of these brands. How many dealerships sell the Suzuki brand but not the Kawasaki or Honda brands?



Answers

① 1,900,800 outcome variations

② 611,226,000 outcome variations

③ 66,679,200 outcome variations

④ $\{0, 4, 6\}$

⑤ $\emptyset$

⑥ $\{-2, -1, 2, 5, 7\}$

⑦ $\{10, 17, 19, 20, 26\}$

⑧ $\{-601, -523, 0, 124, 127\}$

⑨ $\{-16, 1, 3, 6, 7\}$

⑩ i) 8

ii) 32

iii) 54

iv) 7

⑪ i) 19

ii) 6

iii) 32

iv) 6

⑫ i) 17

ii) 10

iii) 42

iv) 14

⑬ i) 53

ii) 18

iii) 4

iv) 19

⑭ i) 15

ii) 7

iii) 23

iv) 18

⑮ 4

⑯ 3